Optimizing Turbine Blade Design

STA 643 — Design and Analysis of Experiments

2026-04-01

Abstract

We present a framework for optimizing gas turbine blade material properties and cooling conditions to minimize maximum von Mises stress, subject to a displacement feasibility constraint ($d < 1.3 \times 10^{-3}$ m) and operational uncertainties on cooling temperature (± 2°C) and pressure load ($\pm 10^4$ Pa). Using all 300 permitted finite-element simulator evaluations — allocated as 100 initial Latin Hypercube Design, 150 sequential Expected Improvement with Constraints, 27 robust grid validation ($3 \times 3 \times 3$), and 23 final validation runs — we achieve a 41% reduction in maximum stress. The GP surrogate models attain 10-fold cross-validated $R^2 > 0.994$. ARD length-scale analysis identifies CTE (Coefficient of Thermal Expansion) as the dominant factor. The recommended design maintains 100% displacement feasibility under operational uncertainty with a 45% safety margin.

1 Introduction

1.1 Background

Gas turbine engines convert high-temperature, high-pressure combustion gases into mechanical energy for aircraft propulsion and power generation. The turbine blades within these engines operate under extreme conditions: inlet temperatures can be as high as 1,500°C, well above the melting point of most metal alloys. This creates severe thermal and mechanical loads that induce stress and deformation in the blades. Prolonged exposure accelerates engine degradation, making turbine blade design a critical engineering challenge.

Given a limited experimental budget, this project aims to find a material design and cooling schedule for the turbine blades that minimizes blade stress and deformation, while respecting a displacement feasibility constraint and accounting for operational uncertainties in cooling temperature and pressure load.

Physical prototyping is expensive and time-consuming, so we use a finite-element model (FEM) computer simulator that predicts the von Mises stress distribution and displacement across the blade surface. The simulator accepts six input variables and returns two scalar outputs: maximum stress and maximum displacement.

Table 1: Input variables for the FEM turbine blade simulator.

Variable	Description	Range	Type
x_1	Young's Modulus	$[2 \times 10^{11},\ 3 \times 10^{11}]$ Pa	Material
x_2	Poisson's Ratio	$[0.1,\ 0.49]$	Material
x_3	CTE	$[5 \times 10^{-6},\ 1.5 \times 10^{-5}]$ K^{-1}	Material
x_4	Thermal Conductivity	$[5,\ 15]$ W/m/K	Material
x_5	Cooling Air Temperature	$[50,\ 350]$ °C	Operational (± 2°C)
x_6	Pressure Load	$[10^5,\ 4.8 \times 10^5]$ Pa	Operational ($\pm 10^4$ Pa)

1.2 Problem Formulation

Let $\mathbf{x}_d = (x_1, x_2, x_3, x_4)$ denote the material parameters that can be precisely controlled, and let x_5^*, x_6^* denote the nominal operational set-points. The operational inputs are subject to uncontrollable perturbations:

$$x_{5,\text{actual}} = x_5^* + \varepsilon_5, \quad \varepsilon_5 \sim \text{Uniform}(-2,\ +2)$$
$$x_{6,\text{actual}} = x_6^* + \varepsilon_6, \quad \varepsilon_6 \sim \text{Uniform}(-10^4,\ +10^4)$$

The constrained robust optimization problem is:

$$\begin{aligned} \min_{\mathbf{x}_d,\, x_5^*,\, x_6^*} \quad & \mathbb{E}_\varepsilon[\text{stress}(\mathbf{x}_d,\ x_5^* + \varepsilon_5,\ x_6^* + \varepsilon_6)] \\ \text{s.t.} \quad & \Pr_\varepsilon[\text{displacement}(\mathbf{x}_d,\ x_5^* + \varepsilon_5,\ x_6^* + \varepsilon_6) < d^*] \geq 1 - \alpha \end{aligned}$$

where $d^* = 1.3 \times 10^{-3}$ m is the displacement failure threshold. In words, we seek the material and operational settings that yield the lowest expected stress averaged over operational noise, while ensuring the probability of exceeding the critical displacement threshold remains acceptably small. The computational budget is limited to 300 simulator evaluations, each taking approximately 80 seconds.

2 Methodology

2.1 Phase 1: Initial Space-Filling Design (100 runs)

We generated a 100-point maximin Latin Hypercube Design (LHD) in the six-dimensional unit hypercube $[0,1]^6$, where each input is scaled via $z_i = (x_i - L_i)/(U_i - L_i)$. The LHD ensures that each input's marginal distribution is uniform: the $[0,1]$ interval is partitioned into $n = 100$ equal strata with exactly one point per stratum. The maximin criterion further optimizes the arrangement by maximizing the minimum pairwise Euclidean distance among all points. We selected the best LHD from 50 random candidates, achieving a minimum pairwise distance of 0.261. With 100 points in 6 dimensions (~17 per variable), this design provides ample data for fitting a smooth surrogate model.

2.2 Phase 2: Gaussian Process Surrogate Model

We fit independent Gaussian Process (GP) models for stress and displacement. GPs are a natural choice for emulating expensive computer experiments: they interpolate deterministic simulator outputs exactly, provide closed-form uncertainty quantification at untested inputs, and integrate seamlessly with Bayesian optimization via the Expected Improvement criterion. A GP defines a distribution over functions: given observed data, it provides a posterior mean $\mu(\mathbf{z})$ (best prediction) and posterior variance $\sigma^2(\mathbf{z})$ (uncertainty) at any untested point:

$$y(\mathbf{z}) \sim \mathcal{GP}(\mu,\ k(\mathbf{z}, \mathbf{z}'))$$

where μ is a constant mean and $k(\mathbf{z}, \mathbf{z}')$ is the covariance kernel. We use the Matérn-5/2 kernel with Automatic Relevance Determination (ARD):

$$k(\mathbf{z}, \mathbf{z}') = \sigma^2 \left(1 + \sqrt{5}\, r + \tfrac{5r^2}{3}\right) \exp(-\sqrt{5}\, r), \quad r = \sqrt{\sum_{j=1}^{6} \frac{(z_j - z_j')^2}{\theta_j^2}}$$

A short length-scale θ_j means the output changes rapidly with input j (high sensitivity); a long θ_j indicates little effect. This provides a built-in variable importance measure. A nugget term ($\sigma_n^2 \sim 10^{-6}$) is added for numerical stability. Since the simulator is deterministic, the GP interpolates exactly through observed data.

All hyperparameters — σ^2, $\theta_1, \ldots, \theta_6$, σ_n^2 — are estimated by maximizing the log marginal likelihood:

$$\log p(\mathbf{y} \mid \mathbf{Z}) = -\frac{1}{2}\mathbf{y}^\top \mathbf{K}^{-1}\mathbf{y} - \frac{1}{2}\log|\mathbf{K}| - \frac{n}{2}\log(2\pi)$$

where $\mathbf{K}$ is the $n \times n$ covariance matrix with $K_{ij} = k(\mathbf{z}_i, \mathbf{z}_j) + \sigma_n^2 \delta_{ij}$. Model accuracy is assessed via 10-fold cross-validation with hyperparameters re-estimated in each fold (see Section 3.2).

2.3 Phase 3: Sequential Constrained Optimization (150 runs)

With a limited budget, we need an acquisition function that balances exploring uncertain regions of the design space with exploiting regions where the GP predicts low stress, while also respecting the displacement constraint. The Expected Improvement with Constraints (EIC) criterion achieves this by multiplying the expected improvement in stress by the probability of feasibility. We sequentially select new evaluation points by maximizing EIC:

$$\mathrm{EIC}(\mathbf{z}) = \mathrm{EI}(\mathbf{z}) \cdot \Pr[\mathrm{displacement}(\mathbf{z}) < d^*]$$

The Expected Improvement for stress minimization is:

$$\mathrm{EI}(\mathbf{z}) = (f^* - \mu_s(\mathbf{z}))\, \Phi(u) + \sigma_s(\mathbf{z})\, \phi(u), \quad u = \frac{f^* - \mu_s(\mathbf{z})}{\sigma_s(\mathbf{z})}$$

where f^* is the best feasible stress found so far, μ_s and σ_s are the GP posterior mean and standard deviation, and Φ, ϕ are the standard normal CDF and PDF. The constraint probability is:

$$\Pr[\text{disp}(\mathbf{z}) < d^*] = \Phi\left(\frac{d^* - \mu_d(\mathbf{z})}{\sigma_d(\mathbf{z})}\right)$$

We generate batches of 15–20 points per iteration with a diversified strategy (EIC on 80,000 random candidates with minimum-distance penalty). Over 150 sequential runs, the best feasible stress improved from 4.46×10^8 to 2.64×10^8 Pa.

2.4 Phase 4: Robust Validation and Final Verification (50 runs)

After 250 evaluations in Phases 1–3, we select 3 top candidates for robust testing. Candidates are chosen from all feasible designs (displacement $< d^*$) by ranking on observed stress. To ensure the candidates represent diverse regions of the near-optimal space rather than clustered duplicates, we impose a minimum pairwise Euclidean distance threshold in the normalized input space: starting from the lowest-stress feasible design, each subsequent candidate must exceed a minimum distance from all previously selected candidates. Each candidate is then evaluated on a 3×3 factorial grid of operational perturbations:

$$x_5 \in \{x_5^* - 2,\ x_5^*,\ x_5^* + 2\}, \quad x_6 \in \{x_6^* - 10^4,\ x_6^*,\ x_6^* + 10^4\}$$

This yields 9 runs per candidate (27 total), covering the extreme corners of the operational tolerance region. We compare the 3 candidates using mean stress ($\mathbb{E}[\text{stress}]$), coefficient of variation (CV), and feasibility rate (fraction of perturbations with displacement $< d^*$). The candidate with the best robust performance — lowest mean stress and 100% displacement feasibility across all perturbations — is selected as the final recommended design.

The remaining 23 runs provide additional verification: random (x_5, x_6) perturbations sampled via a 2D LHD around the recommended design to further confirm robustness under operational uncertainty.

3 Results

3.1 Exploratory Analysis

The initial 100 simulations reveal that stress spans a nearly 4× range $[4.46 \times 10^8,\ 1.71 \times 10^9]$ Pa, and only 43% of designs satisfy the displacement constraint (Figure 1). Stress and displacement are strongly positively correlated ($r = 0.80$), so optimizing for low stress naturally helps satisfy the displacement constraint.

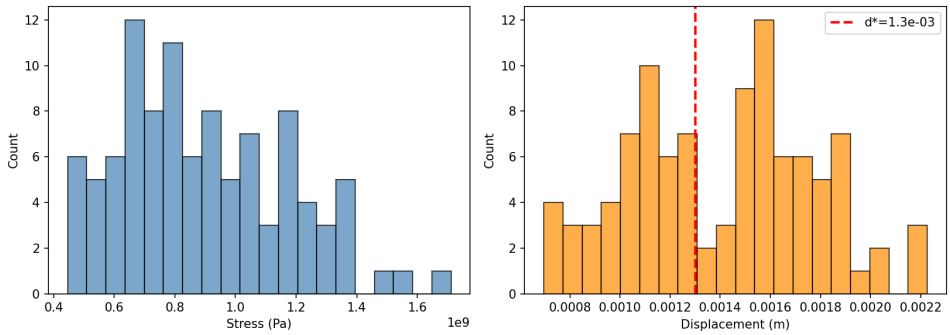

Figure 1: Distributions of stress (a) and displacement (b) from the initial 100-run LHD. Red dashed line: $d^* = 1.3 \times 10^{-3}$.

3.2 Surrogate Model Performance

Both GP surrogates achieve excellent predictive accuracy under 10-fold cross-validation (Table 2; Figure 2), with well-calibrated uncertainty estimates (Figure 3).

Table 2: 10-fold cross-validation results for the GP surrogates.

Response	CV R^2	RMSE	MAPE	95% Coverage
Stress	0.9956	1.82×10^7	1.33%	92%
Displacement	0.9942	2.79×10^{-5}	1.06%	91%

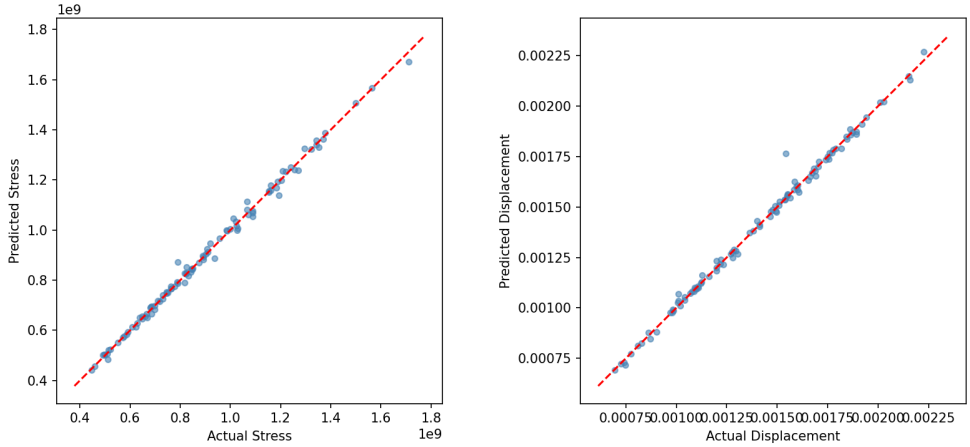

Figure 2: 10-fold CV: predicted vs actual for stress (left) and displacement (right).

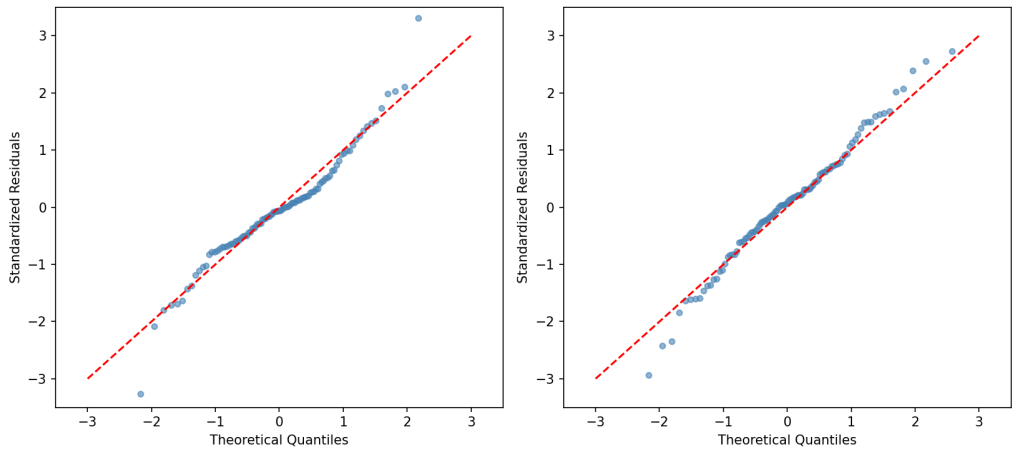

Figure 3: QQ plots of standardized CV residuals confirm well-calibrated uncertainty.

3.3 Optimization Convergence

Table 3: Sequential optimization convergence.

Stage	Cumul. Runs	Best Stress	Improvement
Initial LHD	100	4.461×10^8	—
+ Batch 1 (+20)	120	2.868×10^8	35.7%
+ Batch 2 (+20)	140	2.736×10^8	38.7%
+ Batches 3–8 (+110)	250	2.636×10^8	40.9%

The first 20 sequential runs achieved the largest improvement (35.7%), as the GP rapidly identified the low-CTE corner (Table 3; Figure 4). Subsequent batches refined with diminishing returns, indicating convergence.

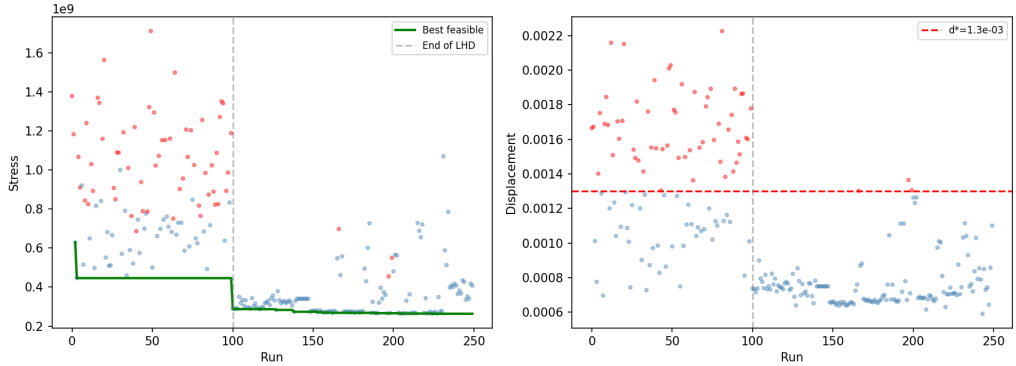

Figure 4: Convergence: (a) stress (blue=feasible, red=infeasible; green=best trajectory). (b) displacement with d^* threshold.

3.4 Robustness and Validation

The three candidate designs selected for robust testing are listed in Table 4. All share the same CTE ($x_3 = 5 \times 10^{-6}$) and cooling temperature ($x_5 = 50°\text{C}$), consistent with the sensitivity analysis, but

differ in Poisson's Ratio, thermal conductivity, and pressure load.

Table 4: Design specifications of the 3 robust-test candidates.

Candidate	x_1 (Pa)	x_2	x_3 (K^{-1})	x_4 (W/m/K)	x_5 (°C)	x_6 (Pa)
C1	2.0×10^{11}	0.477	5.0×10^{-6}	5.00	50	4.049×10^5
C2	2.0×10^{11}	0.469	5.0×10^{-6}	7.93	50	4.395×10^5
C3	2.0×10^{11}	0.476	5.0×10^{-6}	6.47	50	4.350×10^5

All three candidates achieve 100% displacement feasibility across the 3×3 perturbation grid, with Candidate 1 showing the lowest mean stress and smallest variability (CV = 0.36%) (Table 5; Figure 5).

Table 5: Robustness comparison across 3 diverse candidates.

Candidate	$\mathbb{E}$[stress]	Std[stress]	CV	Max disp.	Feasible?
C1 (best)	2.650×10^8	9.47×10^5	0.36%	7.10×10^{-4}	9/9 (100%)
C2	2.697×10^8	1.32×10^6	0.49%	6.57×10^{-4}	9/9 (100%)
C3	2.685×10^8	1.80×10^6	0.67%	6.75×10^{-4}	9/9 (100%)

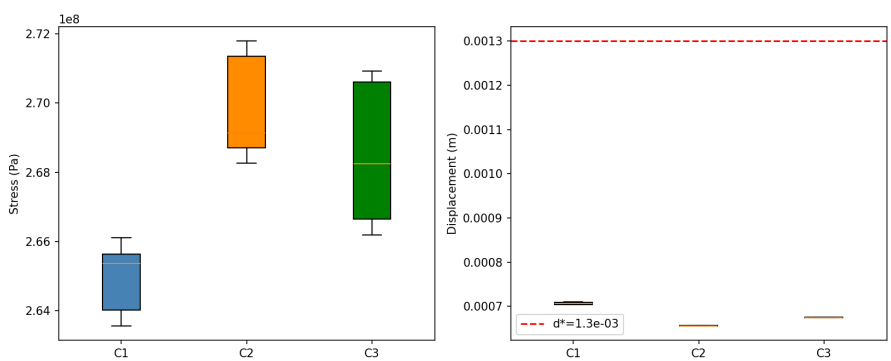

Figure 5: Robust comparison: stress (a) and displacement (b) boxplots for 3 candidates on a 3×3 grid.

The 23 final validation runs further test the recommended design (Candidate 1) under random operational perturbations. All 23 runs satisfy the displacement constraint, with worst-case displacement 45% below d^* (Table 6).

Table 6: Final validation results for the recommended design.

Metric	Value
$\mathbb{E}$[stress] (23 runs)	2.644×10^8 Pa
Std[stress]	7.04×10^5 Pa
CV[stress]	0.27%
Max displacement	7.10×10^{-4} m
Pr(feasible)	23/23 = 100%

Metric	Value
Safety margin vs d^*	45.4%

4 Discussion

4.1 Sensitivity Analysis

The ARD length-scale parameters θ_j from the GP surrogates provide an interpretable variable importance measure. Shorter θ_j means the GP output changes more rapidly with input j. Relative importance is computed as $(1/\theta_j)/\sum_k(1/\theta_k)$.

Table 7: ARD length-scale analysis. For stress, x_3 (CTE) has the shortest θ and highest importance.

Variable	Stress θ	Stress Imp.	Displ. θ	Displ. Imp.
x_1 (Young's Mod.)	5.38	0.177	10.00	0.070
x_2 (Poisson's)	5.57	0.171	1.60	0.440
x_3 (CTE)	**3.16**	**0.301**	5.05	0.139
x_4 (Therm. Cond.)	10.00	0.095	6.74	0.104
x_5 (Cooling)	6.63	0.143	10.00	0.070
x_6 (Pressure)	8.38	0.114	3.97	0.177

CTE (x_3) emerges as the dominant driver for both stress ($|r| = 0.87$, 30.1% relative importance) and displacement ($|r| = 0.90$), as shown in Table 7 and Figure 6. For displacement, x_2 (Poisson's Ratio) also shows high local sensitivity due to nonlinearity.

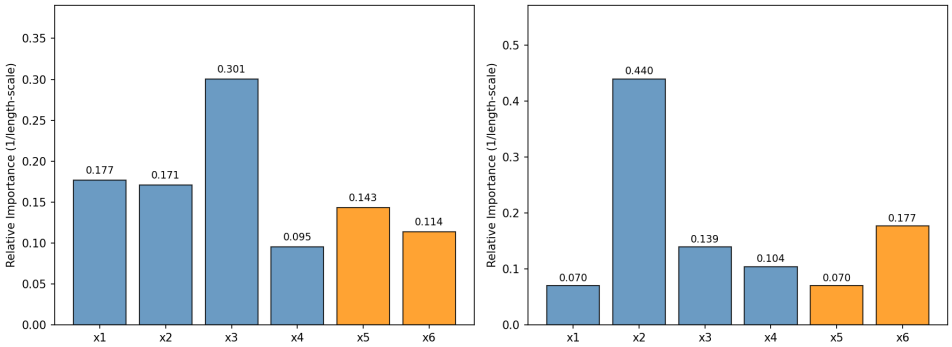

Figure 6: Variable importance from ARD length-scales. Blue: material; orange: operational.

4.2 Recommended Design

Table 8: Recommended blade design specification.

Parameter	Optimal Value	Unit	Note
x_1: Young's Modulus	2.000×10^{11}	Pa	Lower bound
x_2: Poisson's Ratio	0.477	—	Near upper bound
x_3: CTE	5.000×10^{-6}	K^{-1}	**Lower bound**
x_4: Thermal Conductivity	5.0	W/m/K	Lower bound
x_5: Cooling Temperature	50.0	°C	Lower bound
x_6: Pressure Load	4.049×10^5	Pa	Mid-upper range

Consistent with the sensitivity analysis, the optimizer pushes CTE, Young's Modulus, thermal conductivity, and cooling temperature to their lower bounds, while Poisson's Ratio approaches its upper bound to relieve stress concentrations (Table 8; Figure 7). Three variables at their bounds suggests that expanding the design ranges could yield further improvements.

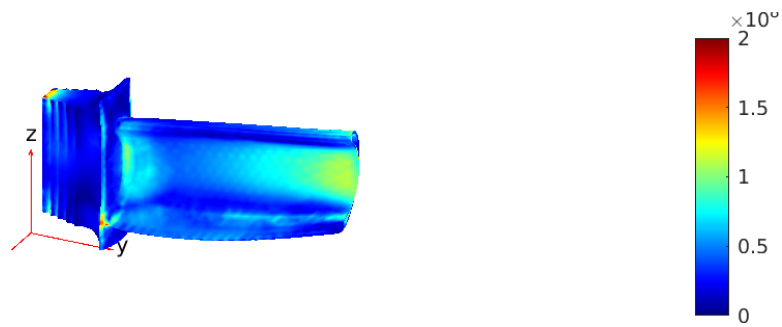

Figure 7: Simulated von Mises stress profile for the recommended blade design. The predominantly blue coloring confirms uniformly low stress ($\sim 2.6 \times 10^8$ Pa).

4.3 Engineering Recommendations

Three recommendations emerge: (1) prioritize alloys with the lowest CTE — it is the dominant factor ($|r| = 0.87$ for stress, 0.90 for displacement, shortest $\theta = 3.16$); (2) operate the cooling system at the lowest achievable temperature; (3) current operational tolerances (± 2°C, $\pm 10^4$ Pa) are safely within the design's robustness envelope (CV $< 1\%$, 100% feasibility).

4.4 Computational Infrastructure

Each FEM simulation takes ~80 seconds, making sequential execution of 300 runs prohibitively slow (~6.7 hours). To accelerate data generation, we built a three-stage SLURM pipeline on the Duke Computing Cluster:

1. **Input generation**: A MATLAB script produces the design matrix (LHD or EIC candidates) and writes it to a shared CSV file (`inputs/params.csv`).
2. **Parallel simulation**: A SLURM array job (`--array=1-N%40`) dispatches up to 40 concurrent tasks, each reading one row from the CSV, calling the FEM simulator, and writing its result to an individual output file.
3. **Result collection**: A final SLURM job (chained via `--dependency=afterok`) merges all individual outputs into a single summary CSV.

The pipeline is orchestrated by a single shell script that chains the three stages via SLURM job dependencies. With 40-way parallelism, a batch of 100 simulations completes in ~4 minutes wall-clock time (vs. ~2.2 hours sequential), a ~33× speedup. This infrastructure enabled rapid iteration through the sequential optimization loop.

5 Conclusion

Through a four-phase framework — maximin LHD, GP surrogate modeling, sequential EIC, and staged robustness validation — we identified a turbine blade design that reduces maximum stress by 41%. The recommended design achieves 100% displacement feasibility under all tested operational perturbations (3×3 grid + 23 random runs) with a 45% safety margin. CTE is the dominant design factor, and prioritizing low-CTE alloys is the single most impactful engineering action. All 300 simulator evaluations were utilized, with the GP (CV $R^2 > 0.994$) enabling efficient convergence in 150 sequential runs.

Appendix

A. Detailed Budget

Phase	Runs	Cumul.	Description
Initial LHD	100	100	Space-filling exploration
Seq. Batches 1–2	40	140	First EIC batches
Seq. Batches 3–8	110	250	Continued optimization
Robust 3×3 Grid	27	277	3 candidates $\times$ 9 perturbations
Final Validation	23	300	23 random perturbations

B. Input–Response Scatter Plots

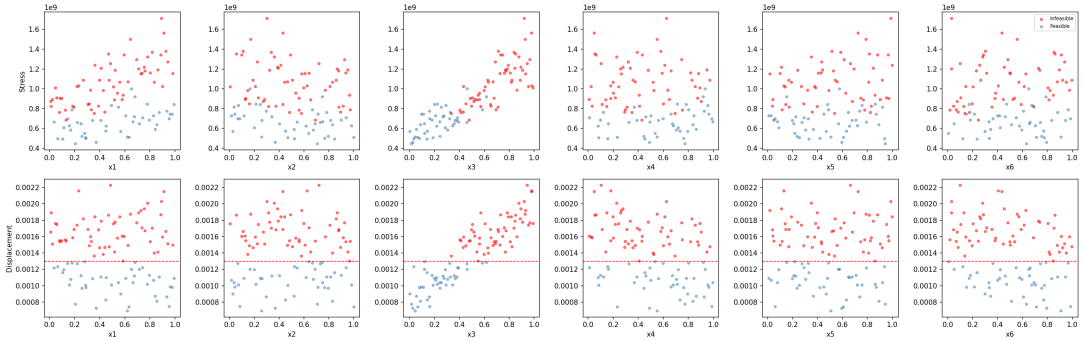

Figure 8: Each input vs. responses (100 LHD). Blue = feasible; red = infeasible.